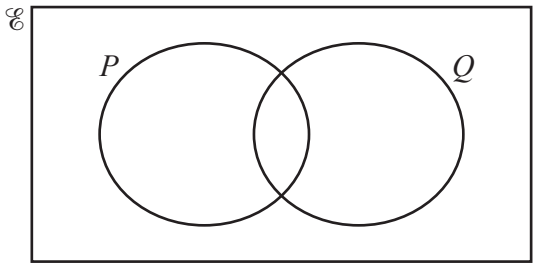
0580_s21_qp_41_Q6

6 **(a)** In the Venn diagram, shade the region $P' \cup Q$.

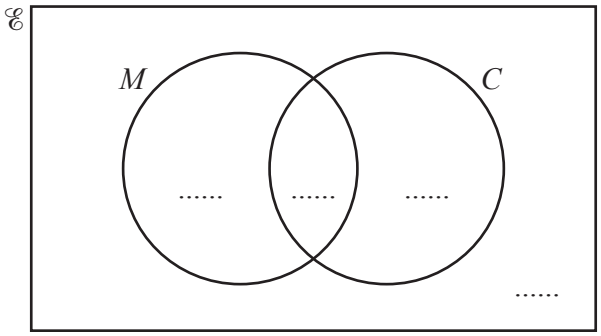


[1]

(b) There are 50 students in a group.

34 have a mobile phone (M).
39 have a computer (C).
5 have no mobile phone and no computer.

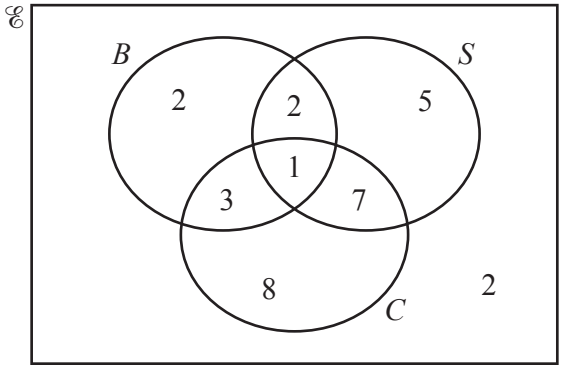
Complete the Venn diagram to show this information.



[2]

0580_s21_qp_41_Q6

(c) The Venn diagram shows the number of students in a group of 30 who have brothers (B), sisters (S) or cousins (C).



(i) Write down the number of students who have brothers.

..... [1]

(ii) Write down the number of students who have cousins but do not have sisters.

..... [1]

(iii) Find $n(B \cup S \cup C)'$.

..... [1]

(iv) Use set notation to describe the set of students who have both cousins and sisters but do not have brothers.

..... [1]

(v) One student is picked at random from the 30 students.

Find the probability that this student has cousins.

..... [1]

(vi) Two students are picked at random from the students who have cousins.

Calculate the probability that both these students have brothers.

..... [3]

(vii) One student is picked at random from the 30 students.

Event A This student has sisters.
Event B This student has cousins but does not have brothers.

Explain why event A and event B are equally likely.

.....

..... [1]